

Endogenous Contestability and Technological Scale: Entry, Selection, and the Value of Comparison

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Abstract

Technological scale can improve production while changing which alternatives enter and what their subsequent performance reveals. This paper develops a Bayesian entry game that separates productive capability from a controllable access wedge. A qualified challenger draws a private setup cost, enters when expected launch revenue covers that cost, and generates an allocation signal relevant to a later adoption decision. The exact decision value is unimodal in the access wedge, even though entry falls monotonically. Accordingly, preserving more entry and preserving more information are distinct objectives. Private and social gatekeeping incentives differ through service surplus, displaced rents, and incomplete appropriation of information. A conditional entry credit implements the efficient cost threshold at a fixed access rule, but does not implement joint gatekeeping and entry by itself. A fixed-cohort reinforced continuation connects these results to finite comparison budgets and a strictly positive Bayesian classification-error floor. The analysis credits established contestability, entry-externality, experimentation, and reinforcement results. Its contribution is the explicit linkage among an entry equilibrium, a consequential decision experiment, and implementable policy margins. The resulting test evaluates qualified alternatives within a declared service boundary while retaining productive scale benefits.

Keywords: contestability, endogenous entry, technological scale, decision value, comparison budget, information externalities.

JEL codes: L11, L13, L41, D83, O33.

1. Introduction

A large technological supplier may produce a better service because it spreads development costs, operates reliable infrastructure, or accumulates useful experience. A smaller hypothetical supplier is not automatically an economically feasible alternative. Nevertheless, the productive technology and the route through which alternatives reach customers are separate objects. A capable incumbent can also control that route. The economic question is whether an actual qualified challenger can enter, whether entry is socially worthwhile, and whether the resulting observations help later decision makers distinguish structural alternatives.

These questions are related but need not have the same answer. A launch can be privately unprofitable yet generate valuable evidence. Conversely, a profitable entrant may largely displace an incumbent without producing enough service or information benefit to justify its setup cost. Most importantly, the probability of observing a challenger does not measure the usefulness of its observation. If a challenger almost always wins under either of two structural states, frequent wins say little about which state holds. A change that reduces entry can, over a range, increase the value of the remaining experiment.

This paper builds a small strategic economy in which these distinctions can be established exactly. An incumbent commits to an access wedge. A technically qualified challenger privately draws a setup cost and enters if expected launch revenue covers it. An uncertain structural state determines the launch probability. The public launch outcome informs a later binary adoption decision with declared payoffs. Private gatekeeping and social welfare are then defined on a common resource boundary. This is a Bayesian game with a solved entry response, rather than a collection of unrelated feasibility inequalities.

The first result derives the access wedge maximizing ex ante decision information, including the probability that the entrant actually appears. The value is single-peaked and its maximizer has a closed form. The second result gives sufficient conditions for a unique gatekeeping equilibrium and identifies the external margin determining its displacement from the welfare optimum. The third derives a credit implementing the efficient entry threshold at a fixed gate. The credit may be negative when entry duplicates costly capability without enough compensating benefit. None of these results prescribes the number of firms from a concentration statistic.

A continuation model then distinguishes a static entry margin from sustained learning. Under fixed candidates, fixed finite profiles, and superlinear reinforcement, eventual monopoly makes the remaining comparison budget finite. Established reinforced-process results and the comparison-budget equivalence theorem imply a positive classification-error floor for a single complete allocation history. This does not show that every industry becomes a monopoly, that new firms can never reveal information, or that endogenous policy changes preserve the theorem's premises. It shows why positive initial entry alone is an incomplete description of informative contestability.

The foundational result is taken from Erlic's revised October 2026 author manuscript, Shadow Futures: Contribution Uncertainty and the Self-Reinforcing Market, associated with the public SSRN record DOI 10.2139/ssrn.6003994. The public record currently describes a July revision; the version used here is the separately supplied October manuscript. The present paper supplies economic decisions and incentives beyond that statistical result. It also makes a limitation explicit: this partial-equilibrium entry game holds the service opportunity, prices, and productive investment fixed unless a particular extension changes them.

The analysis is relevant to platform access, procurement, and technological services where capability is expensive and adoption supplies evidence. An application would need an actual candidate set, a defensible choice model, and measured decision losses. There is no inference here that an existing AI company excludes an efficient rival or deserves fragmentation. Productive scale, costly qualification, and beneficial coordination remain part of the assessment. The object is an implementable comparison route, not entrepreneurial plurality as an end in itself.

2. Related literature and the contribution boundary

Baumol (1982) made potential entry central to industry analysis, showing why realized concentration need not exhaust the competitive discipline of a market. The present game has sunk entry costs and a gate-controlled launch opportunity, so it is deliberately outside a perfectly contestable benchmark. Sutton (1991) explains how endogenous sunk expenditure can sustain concentration as market size grows. Here productive capability and access are separated instead of assigning all concentration to exclusion. These antecedents establish the industrial-organization questions; this paper does not claim to originate them.

Entry is also not a sufficient welfare criterion. Mankiw and Whinston (1986) distinguish private entry returns from social gains when business stealing and setup expenditure matter. Aghion and Bolton (1987) show how contractual arrangements can deter entry by lower-cost suppliers. Our private revenue R , displaced incumbent contribution L , and net service gain m perform different accounting roles. The added decision term is explicitly derived from an experiment rather than appended as an unspecified innovation premium. The resulting credit is conditional on that payoff model and should not be described as a general optimal entry subsidy.

Arthur (1989) studies historical selection under increasing returns. Hayek's competition-as-discovery argument, published in English translation as Hayek (2002), motivates asking what a

competitive process reveals. Neither establishes the statistical claim used here. The allocation continuation is a reinforced stochastic process with specified likelihoods. Oliveira (2009) and Gottfried and Grosskinsky (2024) provide relevant mathematical analysis of dominance and nonlinear feedback. Their established results are credited when deriving the finite-budget continuation, rather than relabeled as new economic impossibility theorems.

Incomplete learning from endogenous observations is already an economic research subject. Rothschild (1974) develops a bandit account of market pricing; Brezzi and Lai (2000) analyze incomplete learning in dynamic allocation. Lai and Robbins (1985) establish efficient adaptive allocation in an informative bandit setting. These papers make clear that actions, sampling incentives, and observation laws jointly matter. Our distinction concerns the entry-weighted value of a particular launch signal and an access decision preceding it. The general possibility that privately chosen actions fail to reveal unknown parameters is not a claimed contribution. Blackwell (1953) gives the appropriate ordering of experiments when the decision problem is held fixed. Blackwell and Dubins (1962) establish merging under absolute continuity, a prediction result that must be distinguished from parameter identification. Bergemann and Morris (2019) organize information design in games, while Bergemann and Bonatti (2019) examine information markets and appropriation. The access wedge here changes both a signal's distribution and whether the experiment occurs. A scalar number of entrants therefore cannot replace a comparison of the full experiments.

Arrow (1962) explains why knowledge production and private appropriation may differ.

Greenwald and Stiglitz (1986) examine externalities with imperfect information and incomplete markets; their general framework is broader than the quasilinear economy below. Jones and Tonetti (2020) analyze nonrival data and private incentives to withhold it. This paper concerns

the production of an informative comparison before a usable data set exists. Nonrival dissemination and trial generation are complementary questions, but sharing a large record cannot restore comparisons that its generating process never supplied.

Dynamic industry models such as Ericson and Pakes (1995) endogenize investment, entry, exit, and industry states. The game below is more restricted and should not substitute for such an equilibrium when those margins are central. Aghion et al. (2005) likewise demonstrate that competition and innovation need not vary monotonically. Our single-peaked formula describes the decision value of one specified experiment, not their innovation mechanism. This literature map locates the proposed additions: an exact weighted signal value, strategic access incentives including that value, and a policy threshold linked to a fixed-cohort information constraint. Priority for these model-specific additions remains to be assessed through a wider literature review.

3. Service boundary, technology, and the Bayesian entry game

Fix the service tasks, customer population, decision horizon, reliability requirement, and resource envelope. Qualification is a technological and organizational condition defined before observed sales. The challenger satisfies it, and its setup cost pays for real preparation under that boundary. Qualification does not assert superiority on every dimension. An alternative requiring a different reliability standard, subsidized input, or incompatible task definition belongs to a different comparison. The unit of policy analysis is consequently narrower than an unspecified industry and need not coincide with a legal market definition.

At productive scale S , the incumbent's service index is $u_I(S)=u_I^0+G(S)$, where G is increasing and concave. This improvement is retained when access changes. For the baseline game S is fixed. The gate variable a lies in $[0,A]$ and shifts allocation log-odds in the incumbent's favor. It

can represent a specified compatibility burden or distribution condition. Increasing a may save support or coordination costs, represented later in payoffs; it is not assumed to be socially wasteful by definition. The real cost of maintaining the gate is also counted.

There are two unknown structural states $\theta \in \{H, L\}$, with equal prior probabilities. Their launch indices satisfy $d_H > d_L$. Conditional on entry, the challenger receives the pivotal opportunity with probability $q_\theta(a)$. The incumbent and challenger know the two indices and prior, but neither observes θ before committing. This is uncertainty about the effect of the service difference in a common choice model, not superior private information held by the gatekeeper. Informational signaling by an informed incumbent would require a different game.

$$q_\theta(a) = \frac{1}{1 + \exp(a - d_\theta)}, \quad \bar{q}(a) = \frac{q_H(a) + q_L(a)}{2}. \quad (1)$$

Timing is as follows. The incumbent commits to a . The challenger then draws a private setup cost F , independently of θ , from a uniform distribution on $[0, \bar{F}]$. The challenger earns private gross return R if awarded the launch and zero otherwise, and pays F upon entry. Assume $0 < R < \bar{F}$, so the uncorrected entry cutoff is interior for every finite a . The launch outcome is observed if entry occurs. Cost realization and entry are observable, but their distribution contains no information about θ . Finally a public decision maker chooses the continuation action using the entry and launch record.

$$F \leq R\bar{q}(a), \quad h(a) = \Pr(\text{entry}) = \frac{R\bar{q}(a)}{\bar{F}}. \quad (2)$$

The cutoff follows from expected private project surplus $R\bar{q} - F$. Since both state probabilities fall strictly with a , entry probability h falls strictly too. An individual cost F is suppressed by an access change from a_0 to $a_1 < a_0$ exactly when $R\bar{q}(a_0) < F \leq R\bar{q}(a_1)$. This is a candidate-specific private viability statement. It neither establishes positive social surplus nor assumes that

lowering a preserves qualification without additional engineering cost. If that engineering cost changes F , the cutoff must be recalculated.

Uniform costs permit an exact equilibrium and convenient closed forms. With a continuous distribution H_F , entry is $H_F(R\bar{q})$, and the expected setup expense is the integral of F up to the cutoff. State-independent costs are essential to the clean signal calculation: if entering firms privately know something about θ , observing entry changes the prior and the experiment must include selection. Similarly, observed performance labels in addition to awards require their own conditional laws. The allocation-only experiment is a declared restriction, not an assertion that actual technological markets reveal nothing else.

4. The exact decision value of entry

The continuation decision is whether to adopt a specified intervention. Action zero pays zero in either state. Action one pays v in state H and $-v$ in state L , where $v > 0$ is a discounted money value external to the launch revenue. Without a signal either action has expected payoff zero. After a challenger award, $q_H > q_L$ makes action one optimal. After an incumbent award, action zero is optimal. If no entry occurs, the prior remains unchanged and the value is zero. The decision problem fixes the action set and payoffs across gates, so changes below genuinely concern evidence.

$$V(a) = \frac{h(a)v[q_H(a) - q_L(a)]}{2} = K[q_H(a)^2 - q_L(a)^2], \quad K = \frac{Rv}{4\bar{F}}. \quad (3)$$

Equation (3) follows by weighting the payoff on a challenger award under each state. It accounts for nonentry and the probability of obtaining each signal. It does not monetize raw transactions. A longer record could be worth more because a decision maker can ignore extra observations, but that record has a different experiment law. The launch value supplies an exactly identified

component of continuation value, not a claim that all future learning is summarized by one award.

Proposition 1. Extend a temporarily to the real line. Let $r = \exp(d_H - d_L) > 1$ and $t = r^{1/3}$. The entry-weighted value V is strictly increasing up to a_V and strictly decreasing after a_V , with

$$a_V = d_L + 2\log t - \log(t + 1). \quad (4)$$

On $[0, A]$, its unique maximizing gate is a_V clipped to that interval. The entry probability remains strictly decreasing throughout. At the unconstrained optimum the state-specific probabilities are $q_L = (t+1)/(t^2+t+1)$ and $q_H = t(t+1)/(t^2+t+1)$.

Proof. Write $q = q_L$ and $q_H = rq/[1 + (r-1)q]$. Since $q_\theta' = -q_\theta(1 - q_\theta)$, differentiation gives

$V' = 2K[f(q_L) - f(q_H)]$, where $f(q) = q^2(1 - q)$. The ratio $f(q_H)/f(q_L)$ equals $r^2/[1 + (r-1)q]^3$. As q increases, q decreases, so this ratio increases strictly from $1/r$ to r^2 . It equals one once, when $q = (r^{2/3} - 1)/(r - 1)$. Substituting t and inverting the logistic probability gives (4). Thus the derivative changes from positive to negative exactly once. Clipping follows from strict monotonicity on either side. Full endpoint and algebra checks appear in Appendix A.

The proposition is a counterexample to a monotone information interpretation of openness.

When the challenger almost certainly wins in both states, a modest wedge makes awards more discriminating. When access is already restrictive, increasing the wedge further reduces both entry and decision value. The result does not recommend exclusion. The gate also affects current service surplus, setup expenditure, and rents, while a different experiment could provide better evidence without restricting launch access. It shows that entry and useful comparison require separate measurement.

For example, if $q_L \geq 2/3$, then both probabilities lie on the decreasing part of f and $V' > 0$. If $q_H \leq 2/3$, both lie on its increasing part and $V' < 0$. These simple sufficient regions are useful in the equilibrium analysis. Neither is a universal property of the logistic family. The conclusion also

depends on the entry response: maximizing $q_H - q_L$ alone gives a different gate because it ignores that the experiment disappears as entry becomes unlikely. A policy counting only surviving trials would miss that extensive margin.

5. Strategic access and the welfare margin

Let $L > 0$ be the incumbent's lost private contribution when the challenger is awarded. Let g_p denote private coordination or support savings, and $c a^2/2$ the real gate-maintenance expense, with $c > 0$. The incumbent captures a fraction $\omega \in [0, 1]$ of the decision value, through its own later decisions or enforceable contracts. Its other productive profit Π_0 is constant at the fixed S . The incumbent anticipates entry, so its expected displacement loss is $L h \bar{q}$. Define $\Lambda = LR/\bar{F}$.

$$\Pi(a) = \Pi_0 + g_p a - \frac{c a^2}{2} - \Lambda \bar{q}(a)^2 + \omega V(a). \quad (5)$$

Social current service surplus changes by m per challenger award, where m is the net real benefit relative to incumbent service, including any displaced productive surplus. It excludes private receipts already counted as transfers. Expected setup expenditure is $R^2 \bar{q}^2 / (2\bar{F})$. Let g_s be total coordination benefits and W_0 baseline social surplus. With $\Omega = (mR - R^2/2)/\bar{F}$, total welfare is

$$W(a) = W_0 + g_s a - \frac{c a^2}{2} + \Omega \bar{q}(a)^2 + V(a), \quad \Omega = \frac{mR - R^2/2}{\bar{F}}. \quad (6)$$

Equations (5) and (6) specify consistent quasilinear payoffs for this partial economy. Launch revenues R and lost private contribution L are not themselves net welfare. Counterpart consumer and incumbent cash flows offset their transfer components; any real service loss is inside m . The information payoff excludes operating effects already inside m and Π_0 . In particular, Ω can be negative when $m < R/2$: private entry may then create negative expected current resource surplus before information is considered. Complete welfare internalization is possible through contracts; $\omega < 1$ is a premise to investigate.

Define the external payoff $E=W-\Pi$, omitting the irrelevant constant. Its derivative is the informative comparison between incentives:

$$E'(a) = g_s - g_p + (\Omega + \Lambda)[\bar{q}(a)^2]' + (1 - \omega)V'(a). \quad (7)$$

Proposition 2. Suppose Π and W are strictly concave and have interior maximizers a_p and a_s . If E' is strictly negative throughout the feasible interval, then $a_p > a_s$. More generally, $W'(a_p) = E'(a_p)$, so a_p is above, equal to, or below a_s according as $E'(a_p)$ is negative, zero, or positive. A sufficient region for $a_p > a_s$ has $g_s \leq g_p$, $\Omega + \Lambda > 0$, and $q_H \leq 2/3$ throughout the interval, with at least one strict inequality contributing to $E' < 0$.

Proof. The challenger enters according to the unique cutoff (2). The incumbent then maximizes the continuous reduced payoff (5), which is unique under strict concavity. Since $\Pi'(a_p) = 0$, equation (7) gives $W'(a_p) = E'(a_p)$. The strictly decreasing derivative of W crosses zero at a_s , proving the order. In the stated region, $[\bar{q}^2]' < 0$ and Proposition 1 gives $V' < 0$, so the derivative comparison holds. This is a Bayesian subgame equilibrium in entry and access, conditional on the commitment timing; it does not assume the challenger observes θ .

Strict concavity need not be asserted without evidence. Appendix A supplies an explicit sufficient curvature bound using the bounded derivatives of a logistic probability. Boundary maxima are covered by one-sided derivative conditions, although the strict ordering statement then becomes a weak comparison. Outside the sufficient region, a private gate can be too low as well as too high. In particular V' can be positive, coordination benefits can accrue to outsiders, or duplicative entry can dominate its current gain. Calling every restriction an external information cost would contradict the model's own first proposition.

For global implementation at fixed entry institutions, a schedule $T(a) = -E(a) + K$ makes the incumbent maximize $W - K$. This identity is a classical corrective-payment application, not a new tax theorem. A marginal charge calibrated at one point only matches first-order conditions there.

Enforcement, financing, participation, and equilibrium responses outside the feasible interval are additional constraints. Aggregate surplus improvement also does not establish an actual Pareto improvement unless losers can be compensated. These limits matter before converting the game into an antitrust or fiscal proposal.

6. Entry correction and productive scale

At a fixed gate, consider a credit s paid upon qualified entry, independent of the launch award. The challenger enters when $F \leq R\bar{q} + s$. The social value of admitting a challenger of cost F is $m\bar{q} - F + v(q_H - q_L)/2$. The marginal launch evidence benefits decision makers regardless of whether it is privately appropriated by the incumbent, while the candidate in this game captures none of it. The policy is conditional on verified qualification and a publicly specified experiment; an unrestricted reward for claiming to be a rival is a different mechanism.

Proposition 3. With nondistortionary financing and the fixed gate, the credit

$$s(a) = (m - R)\bar{q}(a) + \frac{v[q_H(a) - q_L(a)]}{2} \quad (8)$$

implements the efficient entry cutoff $F \leq m\bar{q} + v(q_H - q_L)/2$, clipped to $[0, \bar{F}]$. The efficient entry probability is that clipped cutoff divided by $\bar{F}$. The credit can be negative. It implements a conditional cost threshold, not the joint optimum in gatekeeping and entry.

Proof. Substituting (8) into $R\bar{q} + s - F$ makes private entry surplus equal social incremental surplus. Uniform costs then yield the stated probability. Equivalently, for an admitted fraction h , expected net entry value is $h[m\bar{q} + v(q_H - q_L)/2] - \bar{F}h^2/2$. This is strictly concave in h and its constrained maximizer is the clipped cutoff. A negative optimal s is feasible as an entry payment only if participation and collection permit it. Once s changes entry, the incumbent's displacement term and information receipts change too; equation (6), which used the uncorrected cutoff, must be recomputed.

The proposition separates an entry externality from an access externality. One instrument can address the private setup-cost threshold while leaving the incumbent's prior gate choice distorted. Joint implementation generally needs a gate instrument as well, or a contract adjusting its full anticipated payoff. A candidate's ability to appropriate the continuation benefit would reduce the required credit. Distortionary finance, unverifiable costs, and learning about qualification also alter the mechanism. These are economic constraints rather than reasons to assign the entire social value of competition to every prospective firm.

Productive scale provides a distinct comparative static. Suppose $d_\theta(S) = \beta_\theta[u_E - u_I^0 - G(S)]$, with $\beta_H > \beta_L > 0$, and restrict the relevant S range to $u_E > u_I(S)$, so $d_H > d_L$ remains true. Holding access and launch revenues fixed, $d_\theta'(S) = -\beta_{\theta G}'(S) < 0$. Hence $\partial q_\theta / \partial S = q_\theta(1 - q_\theta)d_\theta' < 0$, and the entry probability falls as the incumbent product improves. This contraction can occur at $a=0$. It is not proof of exclusion: the entrant is confronting a better real alternative.

There is no unconditional sign for $\partial V / \partial S$. Equation (3) gives $2K[q_H \partial q_H / \partial S - q_L \partial q_L / \partial S]$, whose terms depend on both state loadings and current probabilities. An access policy preserving G can therefore have a different information effect from one that changes technology. If S is chosen endogenously, its investment cost and productive profit must enter both private and social objectives, and access conditions may affect incentives to invest. The fixed- S game does not prove that lowering a preserves the privately optimal S . A credible implementation would solve or bound that response.

A simple feasibility benchmark illustrates why even open access can support few qualified suppliers. With M opportunities, equal expected shares among k suppliers, margin $m_0 > 0$, and capability expense $F_0 + fM$ with $F_0, f > 0$, viability requires $k \leq m_0 M / (F_0 + fM)$. The bound approaches m_0 / f as M grows. This is an elementary implication of a stipulated expenditure

schedule, consistent with the question raised by Sutton's endogenous-sunk-cost analysis but not a derivation of his equilibrium. Shared facilities, product differentiation, endogenous prices, or a different expenditure function can remove the bound. Natural oligopoly in this limited sense and modifiable gatekeeping can coexist.

7. Reinforced continuation and decision error

A launch experiment concerns one opportunity. To examine later comparison, condition on entry and freeze the candidate cohort and policy. Let $n \geq 2$ qualified suppliers have finite indices u_i and fixed access weights α_i . Starting stocks $z > 0$ are common across structural parameters. With allocation counts $N_i(t)$ and reinforcement exponent $\rho > 1$, use the recipient probabilities

$$p_{it}(\beta) = \frac{\exp(\beta u_i + \alpha_i)(z + N_i(t))^\rho}{\sum_j \exp(\beta u_j + \alpha_j)(z + N_j(t))^\rho}. \quad (9)$$

Every finite-history recipient probability is positive. The observation contains the allocation history and the declared predictable state; no omitted response channel depends on β . All parameter-specific laws use identical update rules as functions of the realized history. Strong reinforcement in this finite weighted urn leads to eventual monopoly almost surely (Oliveira, 2009; Gottfried and Grosskinsky, 2024). This established absorption property describes adoption feedback, not the productive scale function G . It also does not say that the eventually selected provider has the highest service index.

Let $\varepsilon_t = 1 - \max_i p_{it}$ and $B_T = \sum_{t < T} \varepsilon_t$. After the last allocation to any loser, eventual winner w has count m and the other counts are final. Define C as the sum of loser weights relative to w , including their fixed final reinforcement stocks. Then the residual ε after k more winner allocations is at most $C(z + m + k)^{-\rho}$. Consequently

$$B_{\text{tail}} \leq C \left[(z + m)^{-\rho} + \frac{(z + m)^{1-\rho}}{\rho - 1} \right]. \quad (10)$$

The finite prefix and summable tail give $B_{\infty} < \infty$ almost surely. The last-loser time is a retrospective path property, not an observable stopping rule. The comparison-budget theorem applies because the bounded profile diameter controls ordered-pair conditional separation and the complete-history sigma-field is generated by finite observed histories. It gives mutual absolute continuity of the fixed-parameter laws. Additional parameters affecting unrecorded quality responses or changing candidate sets do not satisfy this allocation-only argument automatically.

Proposition 4. For two distinct admissible fixed parameters H and L satisfying these premises, optimal equal-prior classification from one complete history has strictly positive error R_{∞} . If a correct decision pays v and an incorrect one pays zero, its incremental expected value over the no-signal decision is $v\text{TV}(P_H, P_L)/2$, strictly below $v/2$.

$$R_{\infty} = \frac{1 - \text{TV}(P_H, P_L)}{2} > 0. \quad (11)$$

Proof. The classical binary Bayes-risk identity gives (11). Equivalent probability laws have positive overlapping density mass with respect to their sum, so total variation is strictly less than one. The information-value formula follows by subtracting the no-signal success payoff $v/2$. The result is an economic application of equivalence and a classical testing identity, not a new probabilistic theorem. Appendix B records the finite-horizon bound and explains which expectations are needed for a numerical error floor.

A positive floor need not be large, and one history can still be informative. The impossibility concerns error tending to zero or a nonconstant structural estimand becoming universally consistent as that single history grows. Finite B almost surely does not imply finite expected B . Nor does infinite B imply identification: equal indices can make every recipient law independent

of β . Independent fresh cohorts with a common parameter and an informative finite-block distribution can identify it as the number of cohorts grows. Endogenous arrival of unbounded new firms, redesigned access, and evolving technologies require their own joint model, not an extension by verbal analogy.

8. A decision-specific policy test and empirical implications

The model suggests a policy test that does not take concentration as its primitive. First specify the service and consequential decision. Then identify actual qualified alternatives, their attainable costs, and a modifiable access component. The launch model asks which costs cross a viability threshold while productive quality is held fixed. The welfare model asks whether current service gain plus evidence value exceeds setup and implementation costs. The continuation model asks whether the admitted alternatives remain informative under the ensuing allocation process. These are separate empirical targets.

A first field design would randomize a narrowly specified compatibility or procurement route across independently initialized cohorts. The same service standards, resource budget, and candidate qualifications would apply in each arm. Observe entry decisions and setup spending, award probabilities, quality responses, prices, switching, and reinforcement states. Record exactly which channels reveal the structural state. Outcomes needed to assess service benefit cannot be dropped because they lie outside the convenient recipient likelihood. An estimate of $q_H - q_L$ is useful only when the two structural hypotheses and their empirical interpretation have been specified.

The extensive margin in equation (3) predicts that a policy can improve conditional signal discrimination but worsen ex ante value if it suppresses too many qualified launches. The opposite pattern is also possible. This is testable without asserting that an infinite-horizon budget

is finite. Finite-horizon estimates can compare launch participation, decision losses in a predeclared task, and structural separation under the two routes. A claim of summability needs a validated dynamic mechanism or tail restriction; an observed period of concentration cannot establish it alone.

Identification requires credible variation in the unknown structural effect. In a recommender system the logged assignment policy may be known and independent of that effect. Its probabilities are then not the recipient kernel in (9). Clicks, rewards, or preference labels can be informative even under identical assignments, and their joint likelihood must be modeled. A procurement trial can create useful response variation; simply assigning work at random while observing no parameter-dependent result cannot. This distinction is essential before describing a forced-exploration mandate as a remedy for the stated impossibility.

Independent evaluation may preserve learning without replicating the entire production stack. A shared test facility, compatible interface, or reserved pilot channel can produce informative blocks while retaining scale economies in delivery. Its resource cost and leakage of proprietary returns affect equilibrium incentives. Public release can spread a nonrival signal, but appropriation and strategic use can change behavior. Blackwell's monotonic value concerns a single decision with an option to ignore information; strategic information disclosure requires the separate analysis emphasized by Bergemann and Morris (2019).

The assessment can reject intervention when there is no qualified viable candidate, duplication outweighs likely benefits, or the proposed channel is uninformative. It can support a narrow experiment when social decision losses are consequential and an affordable channel discriminates relevant alternatives. A Herfindahl–Hirschman index may describe the resulting shares, but does not determine any of those premises. Likewise, a smaller nominal provider

count can coexist with better decision evidence, while many providers feeding one common uninformative record can reveal little.

Distribution and institutional feasibility remain material. An efficient entry credit may transfer money toward a challenger and impose losses on an incumbent or customers. Compensation, fiscal cost, enforcement, and incentives to claim qualification need explicit treatment. If the gate provides valuable assurance, removing it changes g_s or setup requirements. A useful next empirical paper would estimate these quantities and solve the resulting constrained intervention problem. The present article establishes the economic distinctions and exact conditional results needed to formulate that study.

9. Conclusion

Technological scale, entry feasibility, and informative comparison are distinct economic margins. The Bayesian game demonstrates that entry falls monotonically with an access wedge while the exact entry-weighted value of a launch signal is single-peaked. Private gatekeeping can therefore differ from the social choice through rents, service effects, and incomplete appropriation, with a direction that depends on the specified state region. A conditional credit corrects the entrant's cost threshold but leaves the prior access choice to a separate instrument.

The fixed-cohort continuation explains why an initially open route can later leave a positive decision-error floor. Its statistical conclusion is conditional on a common allocation experiment and finite comparison budget. The policy object is an affordable, qualified, informative route within a declared service boundary. The analysis supports neither concentration as a sufficient verdict nor more firms as a sufficient remedy. Productive capabilities and the costs of generating evidence belong in the same evaluation.

10. Appendix A: Value and equilibrium calculations

For Proposition 1, set $q=q_L$. Logistic odds imply $q_H/(1-q_H)=r q/(1-q)$, giving $q_H=rq/[1+(r-1)q]$.

Since $1-q_H=(1-q)/[1+(r-1)q]$, division of $q_H^2(1-q_H)$ by $q^2(1-q)$ yields $r^2/[1+(r-1)q]^3$. As q runs from negative to positive infinity, q runs from one to zero. The ratio therefore runs strictly

from $1/r$ to r^2 and crosses one. With $t=r^{1/3}$, the crossing $q=(t^2-1)/(t^3-1)$ simplifies to

$(t+1)/(t^2+t+1)$. Its odds are $(t+1)/t^2$. Equation (4) follows from $\text{logit}(q_L)=d_L-a$. At either infinite endpoint both squared probabilities coincide in their limit, so V tends to zero.

An explicit curvature condition makes Proposition 2 checkable. A logistic probability satisfies

$|q'| \leq 1/4$ and $|q''| \leq 1/(6\sqrt{3})$. The average $\bar{q}$ inherits these bounds. For any such q ,

$|(q^2)''| \leq 2[(1/4)^2 + 1/(6\sqrt{3})] = M$. Equation (3) then gives $|V''| \leq 2KM$. Thus $\Pi'' \leq -c + \Lambda M + 2\omega KM$ and

$W'' \leq -c + |\Omega|M + 2KM$. It is sufficient to impose

$$c > \max\{\Lambda M + 2\omega KM, |\Omega|M + 2KM\}, \quad M = 2 \left[\frac{1}{16} + \frac{1}{6\sqrt{3}} \right]. \quad (12)$$

The derivative bound on q'' follows by maximizing $|q(1-q)(1-2q)|$ on $[0,1]$. Its interior critical probabilities are $(3 \pm \sqrt{3})/6$ and yield $1/(6\sqrt{3})$. The M bound is conservative; using actual indices can establish concavity under a smaller c . Interior solutions additionally require $\Pi'(0) > 0 > \Pi'(A)$ and $W'(0) > 0 > W'(A)$. Otherwise continuous payoffs still attain maxima on the compact interval, and strict concavity still makes them unique, but boundary inequalities replace the interior first-order equalities.

For Proposition 3, a uniform cost cutoff x in $[0, \bar{F}]$ admits fraction $x/\bar{F}$ and incurs expected setup expense $x^2/(2\bar{F})$. Differentiating $h[m\bar{q} + v(q_H - q_L)/2] - \bar{F}h^2/2$ gives the efficient fraction, clipped to $[0,1]$. The credit equates the private and social cutoff, without requiring observation of the actual cost. Incentive compatibility here relies on the candidate bearing its own F and freely choosing

entry. A reimbursement of reported costs would require a separate reporting mechanism and does not follow from the proof.

11. Appendix B: Statistical and replication limits

For (10), loser-to-winner weight after k winner allocations equals $C/(z + m + k)^\rho$, where

$C = \sum_{j \neq w} \exp[\beta(u_j - u_w) + \alpha_j - \alpha_w] (z + N_j)^\rho$ and loser counts are final. Total loser probability is

the ratio divided by one plus the ratio. Since the maximum recipient probability is at least the winner probability, ε is at most total loser probability. The decreasing-series bound

$\sum_{k \geq 0} (z + m + k)^{-\rho} \leq (z + m)^{-\rho} + (z + m)^{1-\rho}/(\rho-1)$ proves (10). The random C is finite pathwise; no finite expectation of it is asserted.

Let D be the profile diameter and β_H, β_L the two fixed coefficients. Along the segment joining them, logit conditional KL can be bounded by a finite constant K_{HL} times ε evaluated at β_H . One permissible constant is $K_{HL} = D^2(\beta_H - \beta_L)^2 \exp(2D|\beta_H - \beta_L|)/2$. The finite-history chain rule gives $KL(P_H^T \| P_L^T) \leq K_{HL} E_H B_T$. Pinsker's inequality therefore bounds equal-prior decision value by

$$V_T \leq \frac{v}{2} \min\left\{1, \sqrt{\frac{K_{HL} E_H B_T}{2}}\right\}. \quad (13)$$

The expectation is indispensable. A finite random budget does not justify substituting a deterministic bound for $E_H B_T$ or taking a finite expectation limit. If a separately established uniform expected bound makes the square root below one, (13) yields a quantitative risk floor. Otherwise equivalence still supplies the qualitative strict floor in (11), while the displayed numerical inequality may be uninformative. Conditional Fisher information and KL separation are not interchangeable with a money value without the decision model.

For the Bayes identity, use $\mu = P_H + P_L$ and densities f_H, f_L . The optimal classifier chooses H where $f_H \geq f_L$. Its error is one half of $\int \min(f_H, f_L) d\mu$, and TV is one minus that overlap integral.

Equivalence means both densities are positive μ -almost everywhere, so the overlap integral is positive. Fixed-law future predictions can nevertheless merge under Blackwell–Dubins. Merging equivalent mixture priors alone would not prove this parameter limitation: mixtures can be equivalent while their fixed-parameter components are mutually singular.

Finally suppose independent fresh cohorts produce N binary launches, each with the same $q_H \neq q_L$ and no shared shocks or hidden drift. Their sample mean converges almost surely to q_θ ; a midpoint classifier identifies θ consistently. Independence, a common state, and distinct block laws are the identifying assumptions. Replication changes the experiment and its resource requirements. It is not contradicted by the single-history theorem, and an unbounded stream of entrants cannot be treated as these independent blocks unless the reset and observation assumptions are established.

12. Statements and declarations

Data availability. This theoretical article uses no observational or experimental data. All results follow from the stated probability laws and payoff functions; there are no empirical estimates. The derivations and parameter restrictions needed to reproduce the calculations are included in the text and appendices.

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References

- Aghion, P., and Bolton, P. (1987). Contracts as a barrier to entry. *American Economic Review*, 77(3), 388–401.
- Aghion, P., Bloom, N., Blundell, R., Griffith, R., and Howitt, P. (2005). Competition and innovation: An inverted-U relationship. *Quarterly Journal of Economics*, 120(2), 701–728. <https://doi.org/10.1162/0033553053970214>
- Arrow, K. J. (1962). Economic welfare and the allocation of resources for invention. In *The Rate and Direction of Inventive Activity: Economic and Social Factors*, pp. 609–626. Princeton University Press.
- Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *Economic Journal*, 99(394), 116–131. <https://doi.org/10.2307/2234208>
- Baumol, W. J. (1982). Contestable markets: An uprising in the theory of industry structure. *American Economic Review*, 72(1), 1–15.
- Bergemann, D., and Bonatti, A. (2019). Markets for information: An introduction. *Annual Review of Economics*, 11, 85–107. <https://doi.org/10.1146/annurev-economics-080315-015439>
- Bergemann, D., and Morris, S. (2019). Information design: A unified perspective. *Journal of Economic Literature*, 57(1), 44–95. <https://doi.org/10.1257/jel.20181489>
- Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2), 265–272. <https://doi.org/10.1214/aoms/1177729032>
- Blackwell, D., and Dubins, L. (1962). Merging of opinions with increasing information. *Annals of Mathematical Statistics*, 33(3), 882–886. <https://doi.org/10.1214/aoms/1177704456>
- Brezzi, M., and Lai, T. L. (2000). Incomplete learning from endogenous data in dynamic allocation. *Econometrica*, 68(6), 1511–1516. <https://doi.org/10.1111/1468-0262.00170>

Ericson, R., and Pakes, A. (1995). Markov-perfect industry dynamics: A framework for empirical work. *Review of Economic Studies*, 62(1), 53–82. <https://doi.org/10.2307/2297841>

Gottfried, T., and Grosskinsky, S. (2024). Asymptotics of generalized Pólya urns with non-linear feedback. *Electronic Journal of Probability*, 29, article 92, 1–56. <https://doi.org/10.1214/24-EJP1157>

Greenwald, B. C., and Stiglitz, J. E. (1986). Externalities in economies with imperfect information and incomplete markets. *Quarterly Journal of Economics*, 101(2), 229–264. <https://doi.org/10.2307/1891114>

Hayek, F. A. (2002). Competition as a discovery procedure. *Quarterly Journal of Austrian Economics*, 5(3), 9–23. English translation of a 1968 lecture.

Jones, C. I., and Tonetti, C. (2020). Nonrivalry and the economics of data. *American Economic Review*, 110(9), 2819–2858. <https://doi.org/10.1257/aer.20191330>

Lai, T. L., and Robbins, H. (1985). Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, 6(1), 4–22. [https://doi.org/10.1016/0196-8858\(85\)90002-8](https://doi.org/10.1016/0196-8858(85)90002-8)

Mankiw, N. G., and Whinston, M. D. (1986). Free entry and social inefficiency. *RAND Journal of Economics*, 17(1), 48–58. <https://doi.org/10.2307/2555627>

Oliveira, R. I. (2009). The onset of dominance in balls-in-bins processes with feedback. *Random Structures & Algorithms*, 34(4), 454–477. <https://doi.org/10.1002/rsa.20261>

Rothschild, M. (1974). A two-armed bandit theory of market pricing. *Journal of Economic Theory*, 9(2), 185–202. [https://doi.org/10.1016/0022-0531\(74\)90066-0](https://doi.org/10.1016/0022-0531(74)90066-0)

Sutton, J. (1991). *Sunk Costs and Market Structure: Price Competition, Advertising, and the Evolution of Concentration*. MIT Press.